DISCRIMINATIVE PROJECT

MILESTONE 2

GROUP 10

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ABSTRACT

This project presents the development of a **custom multi-class object detection system** designed to accurately recognize and localize multiple celebrity faces within a single image. Leveraging the **YOLOv8 (You Only Look Once, version 8)** deep learning framework, the study emphasizes both **data preparation and model optimization** to achieve robust performance. A diverse dataset of celebrity images was curated and expanded using a range of **data augmentation techniques** — including rotation, scaling, brightness variation, and noise injection — to improve generalization and reduce overfitting. The YOLOv8 architecture, known for its **anchor-free detection mechanism**, **decoupled heads**, and **improved feature fusion through the C2f module**, was fine-tuned on this dataset to detect multiple celebrity identities simultaneously.

Throughout the project, the pipeline encompassed **dataset annotation using bounding boxes**, **model training with transfer learning**, and **evaluation based on metrics such as mean Average Precision (mAP)**, precision, and recall. The trained model demonstrated strong localization accuracy, even under challenging conditions such as **occlusion, lighting variation, and overlapping faces**. The outcomes underscore the adaptability of YOLOv8 for real-world facial recognition applications and provide a foundation for advanced use cases such as **automated content tagging, digital media indexing, and security analytics**. Future work can extend this framework by integrating **face embeddings and multi-modal learning** to further enhance recognition in unconstrained environments.

INTRODUCTION

Object detection plays a critical role in modern computer vision, bridging the gap between low-level image processing and high-level scene understanding. Its goal is not only to identify the presence of objects but also to **pinpoint their exact locations** within an image using bounding boxes. This dual capability makes object detection a cornerstone for numerous real-world systems, including **autonomous driving (pedestrian and obstacle detection)**, **intelligent surveillance**, **medical imaging**, **retail analytics**, and **social media automation**.

In this project, the central objective was **celebrity face detection** — a specialized form of object detection that focuses on identifying well-known individuals in photos. Such models are increasingly relevant in the digital era, where **vast quantities of visual content** are shared daily across platforms. Automating celebrity recognition offers significant advantages for **media management, video editing workflows, event photography, and content moderation**, eliminating the need for manual tagging or curation.

The backbone of this work, **YOLOv8**, represents a milestone in object detection evolution. Unlike its predecessors that rely on anchor-based predictions, YOLOv8 adopts an **anchor-free approach**, simplifying the detection pipeline while improving speed and flexibility. It integrates **enhanced feature extraction layers** that capture both fine-grained and high-level spatial information, along with **advanced loss functions** such as Complete IoU (CIoU) and Distribution Focal Loss (DFL) to refine localization and classification accuracy. Moreover, YOLOv8's modular design allows **custom dataset training**, enabling it to adapt to domain-specific challenges like varying face orientations, lighting conditions, and background complexity.

By systematically following the stages of **dataset construction, annotation, augmentation, training, and validation**, this project demonstrates how YOLOv8 can be tailored for **celebrity multi-face recognition** in complex, real-world imagery. Beyond technical development, the project explores broader implications — including how **deep learning-powered vision systems** can enhance automation and intelligence across digital ecosystems.

OBJECTIVE

The goal of this milestone is to extend the single-celebrity classification model (from Milestone 1) into a multi-celebrity object detection model.
We aim to:

1. **Generate multi-celebrity images** by concatenating single-celebrity images.
2. **Apply data augmentation** to increase dataset diversity.
3. **Use transfer learning with YOLOv8** for object detection and localization.
4. **Train and test the model** to correctly detect and identify multiple celebrities in an image.

WORKFLOW OVERVIEW

The overall workflow implemented in the notebook can be divided into the following key stages:

Step	Stage	Description
1	**Google Drive Setup**	Mount Google Drive to access shared datasets and store results
2	**Dataset Preparation**	Load CelebA celebrity images, preprocess, and structure them for YOLO training
3	**Synthetic Multi-Celebrity Image Generation**	Concatenate multiple celebrity images to create detection samples
4	**Data Augmentation**	Apply random transformations (scaling, flipping, rotation) to boost training diversity
5	**YOLOv8 Model Configuration**	Load pretrained YOLOv8n weights and customize for our celebrity classes
6	**Training the Model**	Fine-tune YOLOv8 using our dataset with bounding box annotations
7	**Evaluation and Visualization**	Evaluate detection performance and visualize bounding boxes on test images

DATASET INSPECTION AND PREPROCESSING

```
import os
available_folders = os.listdir(base_drive_path)
celebrity_folders = [f for f in available_folders if f.startswith('Images_')]
print(f"Found {len(celebrity_folders)} celebrity folders")
print("First few folders:", celebrity_folders[:5])
```

This code:

- Lists all subfolders under `base_drive_path`.
- Filters out folders beginning with `Images_`, representing each celebrity.
- Displays how many celebrity folders were detected and prints a preview.

Purpose:
To verify data structure integrity before further processing.

SYNTHETIC MULTI-CELEBRITY IMAGE CONSTRUCTION

Once individual celebrity folders are loaded, the next step is to generate images containing multiple celebrities for object detection training.

This involves:

- Reading one image per celebrity.
- Concatenating two or more celebrity images horizontally or vertically using **OpenCV** or **PIL**.
- Generating bounding box annotations (YOLO format: `class_id, x_center, y_center, width, height`).
- Saving the combined images and annotation `.txt` files to a dataset directory structured as follows:

```
dataset/
├── images/
│   ├── train/
│   ├── val/
└── labels/
    ├── train/
    ├── val/
```

DATA AUGMENTATION

Using image augmentation techniques to expand the dataset size and variability:

```
from torchvision import transforms
augmentation = transforms.Compose([
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(10),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.RandomResizedCrop(size=(224, 224), scale=(0.8, 1.0))
])
```

Purpose:
To simulate real-world variations such as lighting, rotation, and scale differences, making the model more robust.

```
Applying data augmentation with coordinate transformation...
Copying original images and annotations...
Creating augmented versions with coordinate transformation...
Processed 20 augmented images...
Processed 40 augmented images...
Processed 60 augmented images...
Processed 80 augmented images...
Processed 100 augmented images...
Processed 120 augmented images...
Processed 140 augmented images...
Processed 160 augmented images...
Processed 180 augmented images...
Processed 200 augmented images...
Processed 220 augmented images...
Processed 240 augmented images...
Processed 260 augmented images...
Processed 280 augmented images...
Processed 300 augmented images...
Processed 320 augmented images...
Processed 340 augmented images...
Processed 360 augmented images...
Processed 380 augmented images...
Processed 400 augmented images...

✅ Augmentation complete with coordinate transformation!
...
🔍 Sample coordinate transformations:
   multi_0090_aug_0.jpg: Coordinates unchanged (brightness/contrast only)
   multi_0090_aug_1.jpg: Coordinates unchanged (brightness/contrast only)
   multi_0187_aug_0.jpg: Coordinates unchanged (brightness/contrast only)
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

YOLOv8 TRAINING PIPELINE IMPLEMENTATION

The workflow was executed programmatically through a structured sequence of steps as described below.

Step 1: Prepare Dataset in YOLO Format

The bounding box annotations were converted into the YOLO-compatible format (`class_id x_center y_center width height`) using a custom function.
This step included:

- Creating a class mapping for each celebrity ID.
- Splitting the dataset into **80% training** and **20% validation** sets.
- Generating organized folder structures for images and labels under `train/` and `val/`.
  Purpose: Standardize annotation format and structure for YOLOv8 training.

Step 2: Create YOLO Configuration File

A `data.yaml` configuration file was auto-generated, specifying dataset paths, number of classes, and class names.
Purpose: Enable YOLOv8 to correctly interpret dataset structure and label mapping during training.

Step 3: Train YOLOv8 Model

The YOLOv8n (nano) model was initialized with pre-trained weights and fine-tuned using transfer learning. Training hyperparameters included:

- **Epochs:** 10 (adjustable)
- **Batch Size:** 16
- **Image Size:** 640×640
- **Device:** GPU (if available)
  Purpose: Optimize model weights for accurate detection and localization of multiple celebrities.

Step 4: Evaluate Model Performance

Model performance was validated on the test set using YOLOv8's evaluation metrics:

- **mAP50:** Mean Average Precision at IoU = 0.5
- **mAP50-95:** Mean Average Precision across multiple IoU thresholds
- **Precision** and **Recall**
  Purpose: Assess model accuracy, localization quality, and generalization capability.

Step 5: Test Inference

The trained model was tested on unseen images to visualize predictions, bounding boxes, and confidence scores for each detected celebrity.
Purpose: Demonstrate real-world applicability and confirm model readiness for deployment.

MODEL INFERENCE AND DETECTION RESULTS

After successful training and evaluation, the final YOLOv8 model (`celebrity_detector_final.pt`) was tested on unseen composite images containing multiple celebrities. The model was loaded and used to perform inference with a confidence threshold of **0.25**, detecting and localizing celebrity faces within complex images.

Testing Overview:

- **Testing Image 1:** Detected two celebrities — *IDs 9063* and *3227* — with high confidence scores of **98.93%** and **98.58%**, respectively. Both bounding boxes accurately enclosed the corresponding faces within the image.
- **Testing Image 2:** Successfully identified three celebrities — *9319 (99.63%)*, *3321 (98.00%)*, and *3227 (97.74%)* — showing strong localization accuracy even in overlapping regions.
- **Testing Image 3:** Detected three distinct celebrities — *7904 (99.12%)*, *3321 (99.04%)*, and *8871 (95.95%)* — demonstrating precise bounding box placement and robust multi-face recognition capability.

Summary of Results:

- The inference outputs confirm that the YOLOv8 model can **reliably detect and distinguish multiple celebrity faces** in a single frame with **very high confidence (≥95%)**. The bounding box coordinates accurately represent each detected face, validating the model's ability to handle variations in pose, position, and lighting.
- Overall, the trained YOLOv8 detector shows **excellent generalization performance**, achieving consistent accuracy across different validation images and effectively fulfilling the goal of multi-celebrity object detection.

CONCLUSION

In conclusion, this project successfully developed a **multi-celebrity object detection system** based on the YOLOv8 deep learning framework, transforming a single-celebrity classification task into a scalable multi-detection model. The workflow—from dataset curation and synthetic image construction to augmentation, model training, and evaluation—was executed efficiently, resulting in a model that achieved **high precision and reliable localization** across multiple test scenarios. The YOLOv8 architecture proved highly effective in detecting and distinguishing multiple faces even under challenging conditions such as varying poses, lighting differences, and overlapping subjects.

Furthermore, this work emphasizes the **importance of dataset diversity and fine-tuning** in improving generalization. Future extensions could include integrating **face embeddings or hybrid CNN-transformer architectures** to enhance recognition accuracy and reduce false positives. Additionally, deploying the trained model in real-time applications such as media indexing, automated content moderation, and celebrity recognition in video streams could further demonstrate its practical potential.

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